

Project Proposal

Research Question

How can **addictive design principles** be applied towards supporting **hyperfocus** state (increasing attention span) of **neurodivergent** Computer Science students when learning **programming**? (Ashinoff & Abu-Akel, 2019; Hupfeld et al., 2018)

Relevance and Users

In introductory programming courses, many neurodivergent students, such as those with ADHD or autism spectrum conditions, face challenges in maintaining sustained attention during coding exercises. These difficulties can lead to frustration, decreased motivation, and lower learning outcomes (Hupfeld et al., 2018). We want to explore whether existing addictive design principles can support students in keeping attention, and potentially demonstrate their ability to enter hyperfocus states (periods of productive concentration), when engaging with stimulating, and rewarding activities. (Fogg, 2009; Argasinski & Wegrzyn, 2019)

The objective of this project is to explore how **addictive design principles** (Eyal, 2014; Zagal et al., 2013), such as streak systems, level-ups, leaderboards, infinite scroll, and loot boxes, can be intentionally implemented in an educational context to support hyperfocus and improve engagement with learning materials. (Zichermann & Cunningham, 2011; Sailer et al., 2017; García-Redondo et al., 2019)

The target users are **first-year Computer Science students at the University of Copenhagen (KU)** enrolled in the **Programming og Problemløsning (PoP)** course running from September to January.

Through this project, we aim to deliver insights into how gamification and addictive design mechanisms can be ethically repurposed to enhance focus and sustained engagement in.

Investigation Approach

- **Pre-Study Interviews (Late August – Early September)**

Conduct interviews with 3–5 students who have previously taken the PoP course to understand challenges in maintaining focus during programming sessions. Gather insights on motivational triggers, distractions, and potential positive reinforcements for neurodivergent learners.

Meet with the PoP course manager before the semester starts to identify the programming exercises that will be taught in November. This ensures the prototype

content aligns with course material.

- **Theoretical Framework (September – October)**

Review literature on:

- Addictive design principles in digital environments.
 - Gamification in educational technology.
 - Attention management and hyperfocus in neurodivergent learners.
- Build a framework for selecting which addictive mechanisms may ethically enhance focus.
- **Prototype Design and Implementation (October)**
Develop a **web-based gamified coding platform** similar to Duolingo or LeetCode but focused on PoP course material. Include experimental addictive elements (e.g., streaks, leveling up, leaderboards, timed challenges) to test which designs foster hyperfocus without negative side effects.
- **Pilot Testing (November – 2 weeks)**
Recruit 3–5 students for in-depth usability testing with regular feedback sessions. Collect both **qualitative data** (interviews, think-aloud protocols, experience reports) and **quantitative data** (time-on-task, number of completed exercises, re-engagement rates). Optionally, collect broader, less detailed data from additional students using the platform during the test period.
- **Data Analysis and Reflection (Late November – December)**
Analyze results to determine which design principles positively affected attention span and perceived ability to hyperfocus. Reflect on potential ethical considerations of using addictive design in educational tools.

Expected Result

The project is expected to deliver:

- A **web-based coding app prototype** adapted to PoP course content.
- Insights into which addictive design elements most effectively support focus and engagement for neurodivergent learners.
- Recommendations for ethically implementing these principles in future educational tools.
- A **final report** and **oral exam presentation** summarizing findings, methodology, and design recommendations.

Uge 35 (25/8-2025) - Uge 5 (26/1-2026)

Ugestal	Aktiviteter
Uge 35 (25/8-2025)	- Møde med Pernille - Skrive kontrakt
Uge 36 (1/9-2025) – Uge 37 (8/9-2025)	- Artikellæsning - Finde PoP-studerende til vores fokusgruppe
Uge 38 (15/9-2025)	- Pre-interview af vores fokusgruppe - Start med at lave vores produkt - Teoriskrivning
Uge 39 (22/9-2025) – Uge 43 (20/10-2025)	- Udvikler vores produkt - Teoriskrivning
Uge 44 (27/10-2025)	- Eksamensugen
Uge 45 (3/11-2025)	- Re-eksamensugen (ferie)
Uge 46 (10/11-2025)	- Afslutning af produkt - Test af produkt
Uge 47 (17/11-2025)	- Lanceringen af produktet til PoP studerende og fokusgruppen - Observation af produktet under brug / vi vil være tilgængelig hvis der opstår fejl
Uge 48 (24/11-2025)	- Post interview af vores fokusgruppe - Databehandling
Uge 49 (1/12-2025) – Uge 2 (12/1-2026)	- Databehandling - Rapportskrivning
Uge 3 (5/1-2026)	- Aflevering af rapporten
Uge 5 (19/1-2026)	- Mundtlig eksamen

References

- Argasinski, J. k., & Wegrzyn, P. (2019). Affective patterns in serious games. *Future Generation Computer Systems*, Volume(92), 526-538. <https://doi.org/10.1016/j.future.2018.06.013>
- Ashinoff, B. K., & Abu-Akel, A. (2019, 09 20). Hyperfocus: the forgotten frontier of attention. *Psychological Research*, Volume(85), 1-19. <https://doi.org/10.1007/s00426-019-01245-8>
- Eyal, N. (2014). *Hooked: How to Build Habit-Forming Products*. Penguin Group (Canada).
- Fogg, B. J. (2009). A behavior model for persuasive design. *Proceedings of the 4th international Conference on Persuasive Technology*, 1-7. <https://doi.org/10.1145/1541948.1541999>
- Galeos, C., Kapouzis, K., & Tsatiris, G. (2020). Developing an educational programming game for children with ADHD. *International Workshop on Semantic Media Adaptation and Personalization (SMAP)*, 1-6. 10.1109/SMAP49528.2020.9248458
- García-Redondo, P., García, T., Areces, D., Núñez, J. C., & Rodríguez, C. (2019). Serious Games and Their Effect Improving Attention in Students with Learning Disabilities. *International Journal of Environmental Research and Public Health*, 16(14), 2480. <https://doi.org/10.3390/ijerph16142480>
- Hupfeld, K. E., Abagis, T. R., & Shah, P. (2018, 09 298). Living “in the zone”: hyperfocus in adult ADHD. *ADHD Attention Deficit and Hyperactivity Disorders*, Volume(11), 191-208. <https://doi.org/10.1007/s12402-018-0272-y>
- Sailer, M., Hense, J. U., Mayr, S. K., & Mandl, H. (2017). How gamification motivates: An experimental study of the effects of specific game design elements on psychological need satisfaction. *Computers in Human Behavior*, Volume(69), 371-380.
- Zagal, J. P., Björk, S., & Lewis, C. (2013). Dark Patterns in the Design of Games. *International Conference on Foundations of Digital Games*. <https://www.semanticscholar.org/paper/Dark-patterns-in-the-design-of-games-Zagal-Björk/19a241378b06d868eb5f6b76027172c3aaca86f4>

Zichermann, G., & Cunningham, C. (2011). *Gamification by Design: Implementing Game Mechanics in Web and Mobile Apps*. O'Reilly Media.

Pre-study Interview Questions

1. What do you think was the most challenging part of the POP course at the time you took it? Was it related to any specific learning topic?
2. During the class/learning, what are the factors that make you keep or miss your attention the most? How do you think they can be minimised?
3. What do you miss to have more for the POP course such you would feel to ease and enhance your learning of programming? What has worked and what has not?
4. What kind of game are you hooked on in your personal life? Could you elaborate why you are hooked/fascinated by this game? (if not play games, ask with social media)
5. Would a Tamagochi + Pokemon like programming learning game intrigue your attention? Would you be likely to make use of it? What kind of element do you think this programming learning game should include such that it contributes to the learning and keeping attention?